Verified Game Software for a Finger Rehabilitation Device with Modes for Stroke, Parkinson's Disease and Dyslexia

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Abstract: **Recovering finger control after a stroke requires practice, but patients often do not get enough therapy time. Home-based games can provide extra practice, although many do not check the measurements they record. This paper presents software for a finger rehabilitation device with ten game modes. Force Pilot records force measurements that differ in Parkinson's disease, while Syllables is a phonics game for readers with dyslexia. The software was checked in three ways: comparing its measurements with independent references, testing its games against known effects in healthy adults, and timing presses from when force first started to rise. A four-database search found no other finger rehabilitation device that used all three. These checks identified a force conversion error that made readings 4.5 times too high and an audio delay of 87 ms assumed to be 40 ms. On Curtin's EEG laboratory task, all four healthy adults slowed by 91 ms on average when the sequence turned random, showing they had learned it. Presses were counted a median 40 to 55 ms after the finger first started applying force. This delay stayed within one 5 ms sample when the sequence turned random, although it shortened as three participants practised, accounting for 8 to 30 percent of their speed-up. Both EEG recordings showed all 1,859 and 1,794 event markers were within one sample of the logs. One volunteer ran a session alone, and all six blocks were saved. The device is ready for a patient pilot, with a normative study alongside it.**

Keywords: stroke rehabilitation, Parkinson's disease, dyslexia, serious games, force sensing, measurement verification, sequence learning, EEG event markers

I. Introduction

About 12.2 million strokes occur each year worldwide [1], and only about 38 percent of those with a flaccid arm regain some dexterity within six months [2]. Recovery depends on practice. Strong evidence favours intensive, high-repetition, task-specific training [3], and more therapy time gives more improvement [4], yet half of arm sessions lack functional practice and the rest average 32 repetitions [5]. Finger games can add practice at home: MusicGlove proved feasible at home [6], while robots such as FINGER [7] and PRIDE [8] track finger flexion angles but stay in laboratories.

There is still a gap between these two types of devices. Home-based games record performance data, but few studies show it has been checked against an independent reference. This matters because precision grip force remained impaired even in stroke survivors classified as mildly impaired on the Fugl-Meyer scale, and these measurements changed during recovery [9]. To track recovery, a device's measurements need to match a reliable reference.

This paper presents the game software the author wrote for a finger rehabilitation device (Fig. 1) to address this gap, and its testing in healthy adults before patients. It builds on earlier work at Curtin University dating back to 2023 [10], [11], [12]. Patient scores cannot be properly understood without the normal values, the device's measurement error and the improvement from practice alone. Also, a game that reproduces a known effect in healthy adults likely measures what it is intended to measure. For this reason, six of the eight study modes were designed around effects reported in the literature, with each expected result defined before testing began.

The question is whether a finger rehabilitation device can accurately measure finger function, checked against independent references and in healthy adults before a patient pilot. The device needed to meet four requirements: (i) every electroencephalography (EEG) marker within one 2 ms sample of the log; (ii) every study press logged within 50 ms of its threshold crossing; (iii) every participant showing the expected learning effect; and (iv) a participant running a session alone with every block saved. A check added after the data asked whether the delay from a press's force rise to its count changed by more than one 5 ms sample between the two blocks compared for learning. The paper makes two contributions:

- Game software for the device, whose ten game modes share one engine and four force sensors, calibrated to each person's hand, with EEG markers within one sample of the log and two constants corrected against independent references.
- A testing method for other force-based game devices: an independent reference for each measurement, a published effect for each game and press timing from when force first starts to rise. In healthy adults, the shorter delay with practice accounted for 8 to 30 percent of three participants' improvement in speed. All four participants, two with EEG, showed the sequence-learning effect, and a fifth participant ran a session alone, saving every block.

II. Related Work

A. Devices and what they record

Low-cost rehabilitation devices suit home use, but often record limited information that clinicians can use. MusicGlove does not measure individual finger force [6], and virtual reality games were safe in supervised settings, although the supporting evidence is limited [13]. Force-tracking devices measure more, with improvements reported in grip force control in an uncontrolled pilot [14] and hand function [15], and with FINGER [16] and keypad individuation training [17]. However, many cheaper devices report performance data without checking how accurate it is, and good reliability does

not mean accuracy: GripAble showed strong agreement with a Jamar dynamometer, with an intraclass correlation (ICC) of 0.91, but read only 69 percent of the Jamar's force [18].

Two searches ran in PubMed, Europe PMC, IEEE Xplore and OpenAlex in October and November 2026: finger rehabilitation, force and games, then devices or robots with validity or reliability. Abstracts were checked for what each device measured and whether its validity or reliability was reported, and 11 papers were read in full for any timing check. Of the 19 papers identified, only three devices had tested individual finger force measurements. AMBER measured finger strength, speed and memory, with test-retest ICC values of 0.56 to 0.90 in 36 healthy adults [19]. Another finger individuation device reported ICC values of 0.48 to 0.80 for flexion [20], and a fingertip force device's individuation index was related to manual ability in children with cerebral palsy, in the preferred hand only [21]. None of these devices tested their training games against published effects in healthy adults, and none reported checking timing against an independent reference (Table I).

TABLE I. FINGER REHABILITATION DEVICES AGAINST THE CHECKS OF THIS PAPER, FROM THE AVAILABLE REPORTS AND ABSTRACTS (NO: NOT REPORTED).

Device	Force from each finger, every press	Timing checked against a reference	Known healthy effect	EEG markers
PRIDE [8], virtual keypad [17]	No	No	No	No
FINGER [7], force tracking [15], AMBER [19]	Some fingers or tasks	No	No	No
Individuation [20] and fingertip force [21] devices	Yes	No	No	No
Curtin, 2023 to 2025 [10], [11], [12]	Partly	No	Partly	Untested
This device	**Yes**	**Yes**[a]	**Yes**	**Yes**

[a] The laboratory's screen and sound delays are still to be timed (Section VI.D).

B. Validity from known effects

Digital or shorter versions of laboratory tests are usually compared with the original task before use, as for a tablet Corsi test [22], a three-minute vigilance test [23] and a mobile timing battery [24]. Gamified versions mostly give similar results, though game elements can change the abilities tested [25]. In healthy adults, construct validity can be checked by whether a task reproduces established effects. An exponential waiting time before a cue, known as the foreperiod, reduces anticipation, so reaction time (RT) should not decrease as the wait increases [26]. When one finger presses, some force transfers to fingers that should remain still, mostly neighbouring fingers [27]; this enslaving is around 8 to 10 percent of total force when two fingers hold a quarter of their maximum [28]. Each extra finger in a chord adds about 126 ms [29]. In the serial reaction time task, participants respond more slowly when the practised sequence changes to random order [30], and the learning also shows in EEG. An item that breaks the expected sequence produces a larger N2, a negative wave around 200 to 300 ms after the stimulus [31], and the P3, a later positive wave around 300 ms, becomes smaller as targets become more predictable [32]. Beta power (EEG activity at 13 to 30 Hz) over the motor cortex is higher during random trials than learned sequences [33]. People tap slightly before a beat [34], less so to music [24], and young adults release force at least as accurately as they apply it [35]. The chord time rests on a single study and the release ratio on one group's studies, so their expected result is a direction or range, not one value. Section V tests four (sequence learning, beta, N2 and P3) and Section V.D the chord timing and release ratio in single participants; the rest wait on the normative study.

Reliability needs separate checking, as an effect can appear consistently across participants but be poor at telling individuals apart if everyone produces similar results [36]. Reliability is assessed using the intraclass correlation ICC(2,1), with the categories proposed by Koo and Li [37] and its 95 percent confidence interval (CI). The standard error of measurement and minimal detectable change show how much error to expect in the measurement's own units [38]. Terms follow the COSMIN taxonomy [39], and reporting follows the GRRAS guidelines [40]. For EEG, accurate reporting of event timing is important [41], and each laboratory needs to measure its own delays with a photodiode on the screen and a microphone at the speaker [42].

III. THE DEVICE

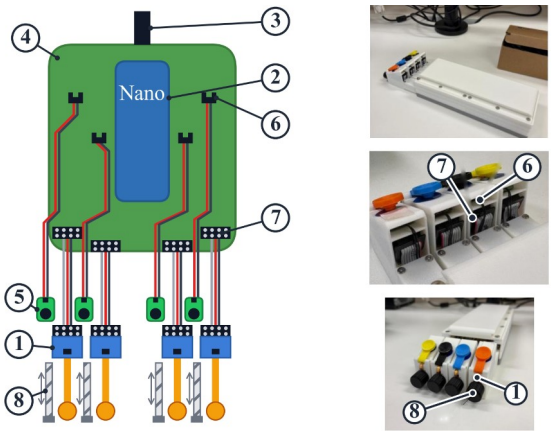


Fig. 1. The finger rehabilitation device, drawn and photographed: 1 SingleTact sensor and interface board, 2 Arduino Nano, 3 USB cable, 4 custom circuit board, 5 motor board, 6 motor board connector, 7 force sensor connector, 8 threaded finger-rest rod.

A. Hardware and press detection

Each finger rests on a factory-calibrated SingleTact capacitive force sensor that measures up to 10 N. Its interface board divides this range into 512 counts, 51.2 counts per newton, within 2 percent of full scale [43], [44]. An Arduino Nano reads all four sensors every 5 ms and drives a vibration motor beneath each finger. Fig. 2 shows one press.

At the start of each session, a short calibration records the sensors unloaded, with the hand resting and during one light press per finger. This gives each finger's difference g between resting and the light press, which sets the thresholds, with a floor. Each raw sample x is smoothed to s, and the baseline b follows s only while the finger is released, starting from its resting value, (1) and (2). The press and release thresholds sit within the calibrated range (3):

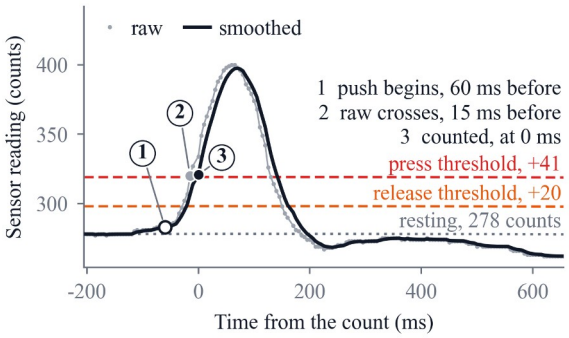


Fig. 2. A slow press replayed through the detector at the board's 5 ms sampling, with raw readings about 255 counts unloaded (press threshold at 40 percent of g; 30 percent in the study): 1 start of the push (Section IV.C), 2 raw threshold crossing, 3 counted press.

$$s = 0.35\,x + 0.65\,s_{\text{prev}} \tag{1}$$

$$b = 0.0005\,s + 0.9995\,b_{\text{prev}} \tag{2}$$

$$T_{\text{on}} = b + 0.30\,g, \quad T_{\text{off}} = b + 0.20\,g \tag{3}$$

A press is counted when s crosses the press threshold, at least 100 ms after that finger's previous event, and released when s falls below the release threshold (Fig. 2). Detection uses sensor counts directly, so its timing does not depend on the counts-to-newtons conversion, and force tracking works in percent of each finger's comfortable maximum, so only forces in newtons depend on the constant. One hand's costed parts come to about AUD 830 to 860, mostly the four calibrated sensors and their boards at AUD 192 each.

B. The ten game modes

All ten modes run on one software engine, designed by the author, which detects presses from the force data and gives feedback through the screen, speakers and vibration motors. For every game, it saves a trial log, the raw 200 Hz force readings and the configuration used; Fig. 3 shows two screens. Two modes reach beyond stroke rehabilitation. Force Pilot measures force characteristics reported to differ in Parkinson's disease [45], [46], and on synthetic signals with a planted 5 percent release error it measured the error to within 0.5 percent of maximum force. Syllables is a phonics-based game for readers with dyslexia [47], with five age-based profiles covering 932 words and audio for every word and syllable. The full session plays eight, not Syllables or two-handed Mirror, each based on a published task: Reaction (choice reaction time), Muscle Memory (serial reaction time), Chords (finger enslaving and chord timing), Rhythm (tapping to music), Force Pilot (visuomotor force tracking), Buzz Hunt (tactile localisation), Echo (Corsi and Simon memory span) and Adaptive (pace based on hit rate).

C. Verification

The device was checked using independent references, simulated testing and an internal design review (Table II). For the audio delay, the laptop's microphone recorded each sound as the software played it; from the play command to the recorded sound, less the operating system's stated, unverified microphone delay, it was 87 ms for the song over five recordings (SD 3.9 ms). Measuring each computer's audio delay is now part of setup. A simulated hand played a whole session through the engine while all 154,035 frames were checked for missing, stale or wrong stimuli. Every fix added a test that fails on the previous code, in a suite of over 4,000 tests covering 87 percent of about 69,000 lines.

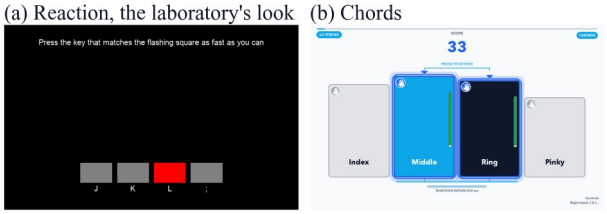


Fig. 3. Two of the ten screens: (a) Reaction in the look of Curtin's EEG laboratory task, its default; (b) Chords, whose finger cards the sequence task also used.

Table III sets out the delays between a physical event and the logged value. On the Arduino's clock, about 95 percent of presses were logged within 16 ms of crossing the threshold on the laboratory computer, which receives each sample separately, and within 36 ms on the development laptop, whose USB adapter delivers samples in bursts 10 to 11 ms late. A reaction time also includes a median 40 to 55 ms from the start of the push to the counted press, plus screen and sensor delays, which cancel between blocks with the same delay, as in the learning score. This delay varies by 7 to 11 ms (SD), against 94 ms (SD) across a participant's trials.

From the manufacturer's specifications, absolute force has a standard uncertainty of about 0.3 N, over half a light 0.5 N press, but similar presses on one sensor differ by about 0.1 N (0.2 N in bench testing [48]), so force is compared within each participant.

D. EEG event markers

With the laboratory configuration loaded, the game engine sends a one-byte event marker on the display frame that presents each stimulus, through a marker box that holds the signal on the EEG amplifier's trigger lines. The device also reproduces Curtin's EEG laboratory PsychoPy serial reaction time task trial by trial, related to the laboratory's published task [49], so the same games can run during a recording.

IV. Checks in Healthy Adults

A. Participants and sessions

Four participants (Persons 1 to 4) completed Curtin's EEG laboratory sequence task on the device's finger cards, shown in Fig. 3(b). All four were invited by the laboratory's head; two were students from the project team, recorded with EEG, and two were laboratory volunteers. Sessions run alone were also offered to the laboratory's healthy student volunteers, on its computer at a time of their choosing. A one-page sheet explained how to adjust the finger rests, log in, choose the session length, calibrate and play, and the software logs show whether participants could complete sessions alone. The full session lasts 45 minutes in one of two fixed orders, A or B, set by a hash of the login name. The first pass plays all eight modes listed in Section III.B. After a three-minute break, Reaction, Rhythm, Force Pilot and Chords are repeated with the same settings to measure within-session reliability, which waits on the normative study.

B. Ethics, inclusion and data

The study was designed to meet the definition of low-risk research under the National Statement on Ethical Conduct in Human Research [50]. The expected risks were discomfort from vibration, light pressing and up to an hour of concentration, and one participant reported discomfort when a sensor stuck (Section V.C). As the invitation came from the

head of their own laboratory, completing sessions alone reduced any feeling of obligation, and the Esc key let them stop at any point. Participants consented by logging in, though the sheet did not state what would be recorded; it now does, and the patient pilot will add a written information sheet and signed consent. The four sequence-task participants agreed beforehand that their data could be recorded, used and retained for this research.

TABLE II. VERIFICATION USING INDEPENDENT REFERENCES, SIMULATED PLAYERS AND THE INTERNAL DESIGN REVIEW.

Check	Found	Now
Sensor datasheet	Force 4.5 times too high	Corrected
Microphone loopback	Audio 87 ms late; 40 ms assumed	Measured during setup
Simulated clock and frame audit	Rhythm cue 9 to 11 ms late; 154,035 frames checked	Cue within 4 ms; no missing, stale or leaked frames
Twelve simulated players	Planted characteristics recovered	*r* of 0.90 or higher; Echo 0.78
Internal design review	16 faults	Each fixed and covered by a test

TABLE III. UNCERTAINTY BUDGET FOR ONE REACTION TIME AND ONE PEAK FORCE MEASUREMENT.

Part	Size	How known
Flash relative to display	At display flip (sequence task); up to one 16.7 ms frame early (Reaction)	Code and frame log
Screen and sensor delays	Sensor under 1 ms; screen untimed	Datasheet; screen to be filmed
Start of push to counted press	Median 40 to 55 ms; interquartile range 10 to 15 ms	Force recordings
Sample to computer over USB	0.8 to 0.9 ms in laboratory; bursts 10 to 11 ms late on development laptop	Arduino clock against computer
Overall reaction time	Systematic 41 to 56 ms (delay and 1 ms timestamp) plus screen and sensor; varies 7 to 11 ms (SD)	Sum, independent spreads
Calibration, linearity and hysteresis	0.12, 0.12 and 0.23 N (2, 2 and 4 percent limits, ÷√3)	Maker; datasheet [51]; dead-weight check pending
Repeatability, resting noise and drift	1 percent of full scale (0.1 N); under 0.03 N	Datasheet and measurements
Overall peak force	About 0.3 N absolute, 0.6 N at *k* = 2; 0.1 to 0.2 N between similar presses on one sensor	Root sum of squares, limits as rectangular bounds [52]

Participants log in with their name to continue an interrupted session and keep the same game order. On import each name is replaced with a code, and the import stops if any name remains, so no name reaches the results or the data. The data are given to the supervisor for permanent storage, following the Australian Code for the Responsible Conduct of Research [53]. The device makes no diagnostic or treatment claims. Its practice is compared with IEC 62304 [54] and ISO 14971 [55] without claiming compliance, and it runs from 5 V USB. Risks a patient's hand adds (tremor, fatigue, vibration, finger-rest rods, hygiene) go to the pilot's clinician.

The patient pilot should also involve Aboriginal and Torres Strait Islander communities from the beginning, because First Nations Australians are hospitalised and die from major heart, stroke and vascular diseases at 1.6 to 2.6 times the age-standardised rates of other Australians [56]. Each cue is a visual tile, an audio tone and a vibration under the finger, so participants with visual field loss still receive it, and a hit now shows a tick as well as its colour. The spoken instructions will still need translating for the pilot. The device fits right hands only, and a failed AUD 192 sensor is replaced alone, not the whole device.

C. Checks and analysis

The laboratory sequence task has 48 random practice trials, eight blocks of a 10-item sequence repeated ten times and 48 random post-test trials. The learning score is the post-test median reaction time minus block 8's, on correct responses under 1000 ms without each block's first trial, as planned. It is also calculated four other ways: against the faster random block; without reversals such as A-B-A, which are slower to answer [57] and twice as common in random trials; without presses under 100 ms, started before the stimulus could be seen; and from the start of the push rather than the counted press, where a push onset was found. Afterwards each participant typed the sequence they remembered, scored by correct triplets against 10,000 random recalls [58]. For Force Pilot, the release-to-press error ratio is the tracking error while force decreases divided by the error while it increases.

The planned checks, statistics and exclusion rules were documented before testing, in a design document public since 2 September 2026 [59], and later changes were dated in the design record. Each mode has its own predefined checks, eleven of which can fail. With eight participants, these will be tested with Wilcoxon signed-rank tests and Holm's correction [60], and reliability with ICC(2,1) [61] and the minimal detectable change at 95 percent confidence (MDC95), the smallest change in one person's score that counts as real rather than measurement error [38]. With fewer than eight participants, individual results are compared with each check's expected range.

The laboratory also asked for a press-timing check: the delay from the start of each push, found in the force signal by the software's onset detector [62], to the counted press. As both come from the same sensors, it excludes the display delay. It looks for a delay that changes as a participant speeds up, bending the recorded learning curve away from the movement, and for a light touch before the push counted as the response.

The four participants' learning scores were analysed with a one-sample t-test, individual bootstrap intervals and a linear mixed model fitted by restricted maximum likelihood, giving each participant their own speed and learning effect [63], with post-test comparisons on three degrees of freedom. Both EEG recordings were average-referenced, filtered at 0.1 to 40 Hz and screened for eye artefacts by independent component analysis, with settings fixed after the first recording. Epochs had a 200 ms baseline and were dropped above 150 µV. On correct responses, N2 was measured at 200 to 300 ms over scalp electrodes FCz and Cz, P3 at 300 to 450 ms over Pz and CPz, and beta power at C3 in the 0.5 s after each stimulus. Each was compared between random trials and late sequence blocks 7 and 8. The two recordings were combined by shuffling trial labels within each participant 10,000 times, with Holm's correction across N2 and P3. The t-test, mixed model, press-timing check, P3 and beta checks, and extra sequence-task readings were added after the data, each dated in the design record.

V. Results

A. Who took part

Six right-handed adults aged 21 to 25 used the device. Four (Persons 1 to 4) completed the laboratory sequence task, and Person 1 also a Buzz Hunt block. Of the ten volunteers sought, one (Person 5) ran a session alone, choosing 30 minutes and playing six of its eight modes in 13.8 minutes of play before stopping. Person 6 completed one Force Pilot block. Every block Person 5 started was saved with its force data, but the first sensor sample took 4.2 s against a 2 s target (1.75 s now, in simulation).

B. Learning on the laboratory's task

All four participants became faster as they practised the repeated sequence and slowed when it turned random (Fig. 4). Their learning scores were 81, 101, 55 and 127 ms, all positive, although Person 3's bootstrap interval extended down to 1 ms. The average was 91 ms (95 percent CI 42 to 140; $t(3) = 5.96$, $p = 0.009$, dz = 3.0), a t-test chosen after the data.

Table IV shows the scores calculated four other ways, and every participant stayed positive in all four. The faster random block helps rule out practice and fatigue. Of these, only Person 3's interval without reversals fell below zero, to −24 ms. The mixed model gave 99 ms (95 percent CI 38 to 160, $p = 0.014$) across 2,984 retained trials. Assuming one shared learning effect gave a misleadingly narrow interval, 82 to 118 ms.

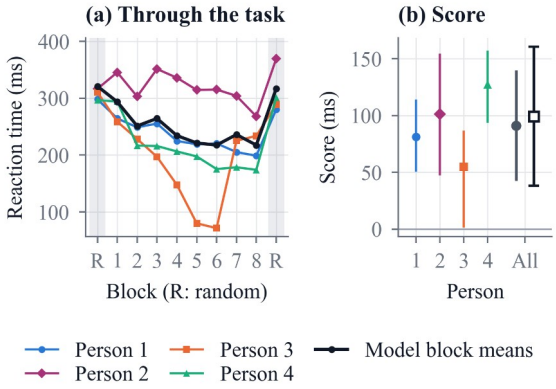


Fig. 4. Sequence learning in four participants. (a) Median RT for each participant (colours) and the mixed model's block means (black), random blocks shaded; Person 3's drop is presses started before the stimulus could be seen. (b) Learning scores with 95 percent confidence intervals; All: the average with its t-test interval (grey) and the mixed model result (black square).

Two participants typed out the whole sequence, and all four recalled more correct triplets than chance ($p = 0.029$ or lower) [58]. Presses made close to the stimulus showed the same: in the sequence blocks, 68 to 94 percent of presses within 100 ms either side of it were on the correct finger, against a one-in-three chance for a guess avoiding the previous finger ($p = 0.002$ or lower). During the random blocks, only 2 of 375 responses occurred this early, and neither was correct. Post-test error rates, 15 to 21 percent, were higher in every participant than block 8's (5 to 21) and random practice's (2 to 9), closest for Person 3 (21.3 against 20.7 percent in block 8), so the slowing was not traded for accuracy.

C. Each press against its push

The push-to-count delay became shorter with practice for three participants, but stayed stable between the two blocks of the learning score. The median delays were 45, 50, 40 and 55 ms for Persons 1 to 4, while force is rising, after the muscles have fired [64]. Each participant's median delays across blocks varied by 10 to 15 ms, but from block 8 to the random post-test they stayed within one 5 ms sample, at 0, +5, 0 and 0 ms (Fig. 5). Reaction times from the push and from the counted press correlated at $r = 0.984$ to 0.997, and learning scores from push onset were within 10 ms of those at the counted press (Table IV). With the onset detector's velocity threshold at 10 or 30 percent of peak instead of 18, the delays moved 5 to 10 ms but the scores at most 4 ms. Light touches were uncommon: no press was released within 60 ms, and only 2 of 15 re-presses within 0.3 s began with one.

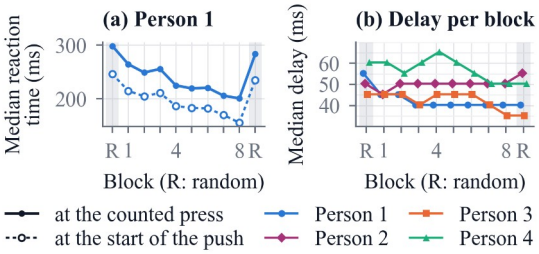


Fig. 5. Each press against its push: (a) Person 1's median RT per block at the counted press (solid) and the push's start (dotted); (b) each participant's median push-to-count delay per block.

One problem appeared: from block 7, Person 3's middle-finger sensor stayed above its release threshold, so further presses on it went uncounted until it lifted. New resting and holding rules release a finger and reset its baseline when it rests below half its peak force for 1.5 s or remains pressed for 2 s. Replayed over Person 3's force data, they recovered presses for all 11 missed middle-finger cues and changed no more than six presses per thousand for the others. The updated learning scores were 81, 104, 55 (95 percent CI 5 to 88) and 127 ms, an average of 92 ms ($p = 0.010$). Without Person 3's middle-finger trials, their score was 45 ms (95 percent CI −19 to 77) and the group average 89 ms ($p = 0.015$), so the group learning effect held despite the sensor problem.

D. The device's own games

Person 5's results were within the expected range for 13 of the 16 checks one session can measure, including 4 of the 6 that could fail, and the single blocks of Persons 6 and 1 for all seven of theirs. The checks outside their ranges were Force Pilot's irregular waves, Buzz Hunt's catch trials and the ring finger in Chords. In Force Pilot, Persons 5 and 6 recorded tracking errors of 2.0 and 3.0 percent of maximum force, 56 and 33 percent below the 4.5 percent scored by holding the best constant force. Their release-to-press error ratios were 1.05 and 0.96, against RMS-error ratios of 0.83 to 0.90 reported in young adults [35], [46]. In Chords, each additional finger added 36 ms, against about 126 ms on keyboards [29], possibly because the fingers already rest on the sensors.

TABLE IV. LEARNING SCORES (MS) CALCULATED FIVE WAYS (SECTIONS IV.C AND V.B); ON THE PLANNED RULE, INDIVIDUAL BOOTSTRAP INTERVALS AND THE GROUP T-TEST INTERVAL; P VALUES FROM T-TESTS ACROSS THE FOUR PARTICIPANTS, CHOSEN AFTER THE DATA.

Person	Planned rule (95 percent CI)	Faster random block	No reversals	No presses under 100 ms	At the push
Person 1	81 (50 to 114)	81	54	64	80
Person 2	101 (47 to 154)	48	65	78	97
Person 3	55 (1 to 87)	55	25	43	47
Person 4	127 (93 to 157)	123	118	127	137
Mean	91 (42 to 140)	77	66	78	90
p	0.009	0.020	0.044	0.022	0.017

E. EEG laboratory sessions

Every EEG event marker matched the game log within one sample: all 1,859 markers from Person 1 and all 1,794 from Person 2, who played the sequence task on 8 and 22 October (BioSemi ActiveTwo, 64 channels at 512 Hz). The first visual response peaked at 121 and 117 ms in random trials and kept its timing through each recording.

Combined, both primary EEG tests passed (Table V; Holm-corrected $p = 0.038$ and 0.006), though the N2 effect came from Person 1 alone. Beta power over the left motor cortex was higher in random trials than the learned sequence in both participants, by 1.1 and 1.3 dB (combined $p < 0.001$), a secondary test consistent with Lum et al. [33]. It was already present in the 400 ms before the stimulus (0.6 and 1.6 dB), so it relates more to being in a random block than to each stimulus. Random post-test responses were also 75 to 81 ms slower than in blocks 7 and 8, which could affect beta, yet beta stayed higher with reaction times matched and before the press (Table V).

Person 2's recording, kept separate and analysed with the settings fixed on Person 1's, kept the beta difference ($p = 0.033$) but not the N2 difference. P3 amplitude fell by 2.4 and 2.5 μV but stayed small in the random post-test, so time on the task could explain the drop as well as learning (Table V).

VI. DISCUSSION

A. What the checks show

All four requirements of Section I were met, as was the press-timing check added after the data, with every marker within one sample, every self-run block saved and every study press logged within about 49 ms of its threshold crossing. All four participants showed the expected sequence-learning effect, although Person 3's interval crossed zero without the reversal or stuck-sensor trials. The average effect was similar to that reported after longer training: 91 and 111 ms after 1,152 trials [58], against 91 ms after 800 here. The device also showed learning in the fingers chosen, not just in reaction time.

B. From a healthy baseline to patients

The normative study, run alongside the patient pilot, will give the normal range in healthy adults, the measurement error and the improvement from practice alone that a patient's score is read against. Of the COSMIN properties [39], the results give initial evidence of construct validity and measurement error (a standard error of measurement of about 20 ms at the push in four participants). Reliability (ICC 0.50, 95 percent CI −0.67 to 0.96), responsiveness and criterion validity wait on the normative study and patient pilot.

TABLE V. EEG RESULTS OF THE TWO RECORDINGS, P VALUES IN BRACKETS; BOTH SHUFFLES LABELS WITHIN EACH PARTICIPANT; LATE: SEQUENCE BLOCKS 7 AND 8; CHECK: ADDED AFTER THE DATA.

Measure	Person 1	Person 2	Both
N2, random post-test minus late, μV [31]	−4.2 (0.020)	−0.4 (0.740)	−2.3 (0.038; Holm 0.038)
P3, random practice minus late, μV [32]	+2.4 (0.023)	+2.5 (0.048)	+2.5 (0.003; Holm 0.006)
P3, random post-test minus late (check), μV	+0.5 (0.644)	−0.5 (0.758)	0.0 (0.973)
Beta at C3, random post-test minus late, dB [33]	+1.1 (<0.001)	+1.3 (0.033)	+1.2 (<0.001)
Beta at C3, learned trials matched in RT (check), dB	+0.8 (0.038)	+1.2 (0.126)	+1.0 (0.020)
Beta at C3 before the press (check), dB	+0.7 (0.010)	+1.4 (0.012)	+1.1 (<0.001)

Each part of the method has support in previous research (Sections II and IV.C), but no device in Table I checks independent references, known healthy effects and press timing together. Other force-based rehabilitation games could use it in three steps: check each measurement against something outside the software, such as a datasheet, microphone recording or EEG system; test each game against an effect reported in healthy participants; and time each press from the beginning of force application, stating how slowly a participant can press before the detector fails.

Because every game shares the engine, logging and checks, the Parkinson's and dyslexia modes needed no new hardware, and other conditions can be added given an established effect to check them. Each session also leaves a per-press record for remote clinical review.

C. Lessons from the engineering

The biggest errors were the force conversion constant and the audio delay. Both were coded as intended and passed their unit tests, yet the force readings were 4.5 times too high and the assumed delay put the laptop's rhythm scores off by 47 ms; each showed only against the datasheet or a microphone. Of 41 faults that could have changed a study result or a session, none was caught by an existing test: 14 showed when the software was run and measured, 2 against the datasheet and microphone, 16 in the design review of each mode (Table II) and 9 in later checks. A unit test only confirms a constant matches the value the author typed, so each one needs an outside reference before participants.

The press-timing check also matters for patients: the count came a median 5 to 10 ms after the raw force crossed its threshold on real presses, but in simulation 11 to 15 ms for a 400 ms force rise and 60 to 86 ms for a 3 s rise, as the resting baseline crept up during a slow press. Where the push onset was found, the shorter delay with practice made up 8, 30 and 13 percent of the block 1 to 8 improvement for Persons 1, 3 and 4, so part of the recorded improvement came from detection rather than movement. For patients, reaction times should therefore be measured from the start of the force rise. On replay, a new two-sample rule counts a press one sample after the raw crossing; on the measured sensor noise, it detected every weak or slow press up to 3 s and no 4 to 6 Hz resting tremor up to 0.3 N. These results check the detector but do not replace patient testing: a sine wave is not a tremor, nor a ramp a slow press. On the sensors, a published effect kept its direction but changed size (Chords, Section V.D), so the normative study must set ranges from the device's own data. Looking back, the checks would have been planned from the

first week and every constant verified before the first mode; two booked afternoons could have given eight participants a session each, but growing from four to ten modes took much of weeks 2 to 8.

D. Limitations

A press counts when the smoothed force crosses its threshold, a median 40 to 55 ms after force began, so reaction times at the count compare with force-sensor studies, not keyboards. The laboratory computer's screen and audio delays are unmeasured, a possible offset common to every marker; the tools for them recovered planted delays to within 1.5 ms in simulation. Presses on a finger resting just above its release threshold, likelier in patients, were missed until it lifted; Section V.C's rules recovered them on replay but are untested in use. The re-press check only sees a light touch if the finger lifts before the push; otherwise the touch is counted as the press.

All participants were healthy young adults, so the results say nothing yet about sensitivity to stroke, Parkinson's disease or dyslexia, and the two project team members' EEG recordings speak only for them. Participants remembered much of the sequence, so some learning was explicit (aware). Knowing only which finger most likely follows the last would get 60 percent of early presses right, not one in three; three participants beat this significantly, but Person 2, with 13 of 19 right, did not. Only one of ten volunteers ran a session, so the normative study will book its participants.

VII. Conclusion

The device met the four requirements of Section I. Two incorrect constants were corrected using independent references, all 1,859 and 1,794 EEG markers matched the game logs within one sample, and all four participants showed the learning effect, 91 ms on average. The device is now ready for a patient pilot, with a normative study alongside it. With individual intervals of about 30 to 54 ms either side (Table IV), one patient session could tell a score larger than this from zero, but not a smaller one. Setting up the pilot includes three checks: the laboratory screen filmed at 240 frames per second, its audio by loopback and each sensor with dead weights. Agreement between passes waits on the normative study, booked with at least eight participants, and for Parkinson's disease, Force Pilot's release measures will be compared with age-matched healthy controls. Devices that score learning at a force threshold should time reactions from force onset or report each block's delay beside the learning results.

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